NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

Project Title: AI-Based Smart Quiz System.

Course Code: CSE 4204
Course Title: Mobile Computing Lab

Submitted By:

Team Name	**:** CSE4204-8A-T03
Section	**:** 8A

GitHub Repository: https://github.com/sudipto224/AI-based-Smart-Quiz-System

Submitted to:

Md. Riaz Mahmud
Designation: Assistant Professor
Department of CSE

Date of Submission: 13/08/2026

TEAM INFORMATION

Official Team Name	CSE4204-8A-T03
Section	8A
Project Title	AI-Based Smart Quiz System
GitHub Repository	https://github.com/sudipto224/AI-based-Smart-Quiz-System

Team Information:

Role	Full Name	Student ID
Team Leader + Backend	Sudipto Das	11220320830
Artificial Intelligence	Rakibur Rahman Mishan	11220320829
Database + Frontend	Aditi Roy	11220320846

Selected AI Platform

The AI-Based Smart Quiz System uses Groq Cloud API as its selected AI platform. Groq provides fast cloud-based inference and supports modern large language models through an API that is straightforward to integrate with a Laravel backend.

Reasons for Selecting Groq

- Free tier offering up to 14,400 requests per day, which is suitable for development and academic testing.
- Fast response time, including sub-100 ms inference performance in supported scenarios.
- Access to the Llama 3.3 70B model for high-quality language understanding and generation.
- OpenAI-compatible API design, which simplifies integration and reduces implementation complexity.
- Easy backend integration with the existing Laravel application and structured JSON-based communication.

AI Model Used

The system uses the Llama 3.3 70B model developed by Meta and accessed through the Groq platform. The model is used primarily for generating educational multiple-choice questions from teacher-provided topics.

Item	**Details**
Model	Llama 3.3 70B
Developer	Meta
Provider	Groq
Primary Use	AI-assisted multiple-choice question generation
Capabilities	High-quality natural language understanding, structured JSON output, MCQ generation with explanations, and low-latency responses.

Why AI is Required

Creating high-quality multiple-choice questions manually is time-consuming, especially when teachers need course-specific and topic-specific questions, plausible distractors, correct answers, and clear explanations. AI is required in the AI-Based Smart Quiz System to automate the first draft of this process. By combining the currently selected course with a teacher-provided topic, the system can generate a structured MCQ for that specific academic context, reducing question-preparation time and allowing teachers to focus on review. The teacher remains in control because every generated question can be reviewed and edited before it is saved.

AI Features Implemented

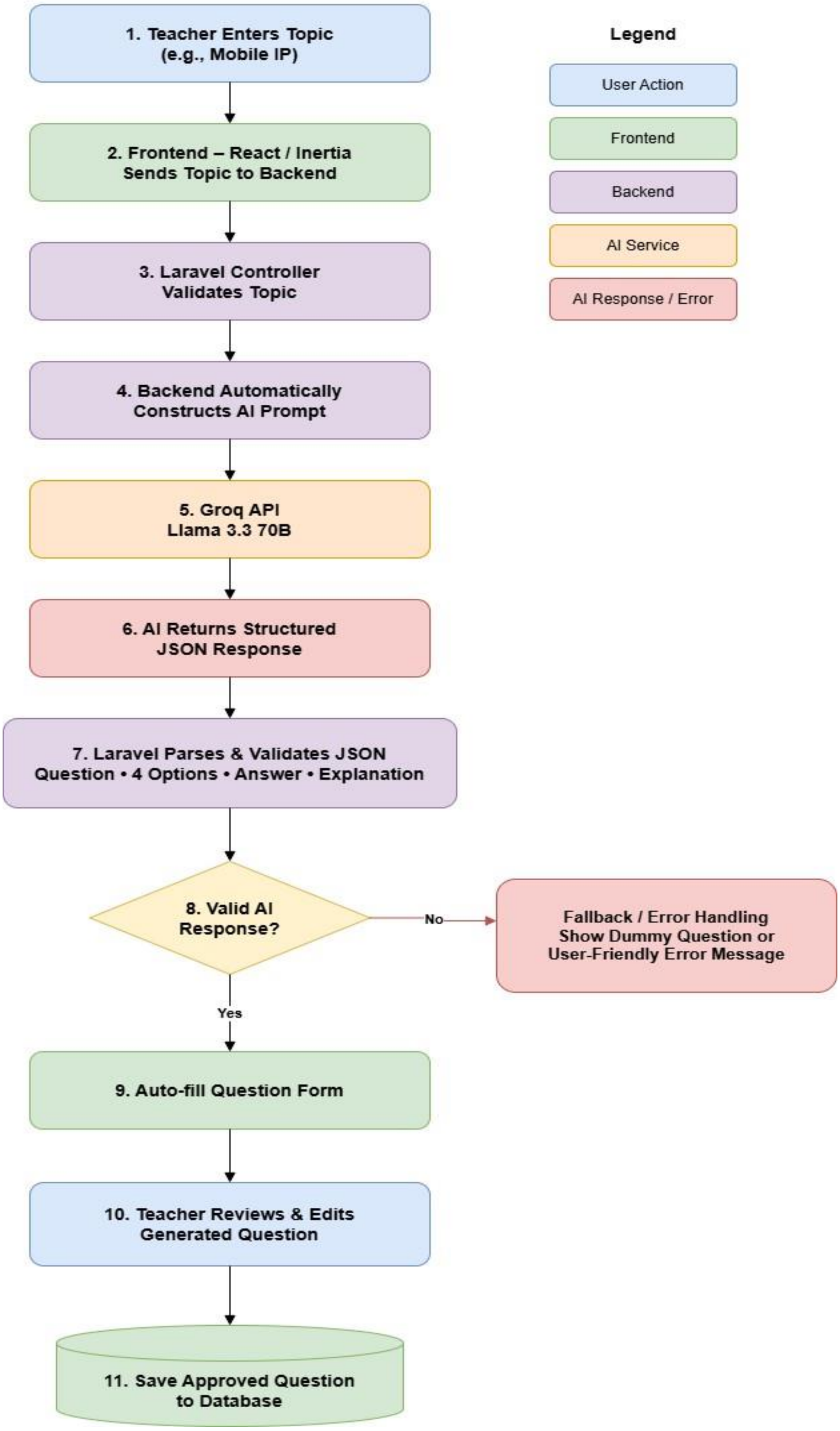
The project currently includes the following intelligent features. The first feature directly uses a large language model, while the second uses a rule-based heuristic to identify suspicious quiz behaviour.

Feature	Implementation	AI Type
AI Question Generation	Teachers open a course and enter a topic. The Laravel backend automatically combines the selected course name and topic, then the Groq API generates a course-specific MCQ containing a question, four options, the correct answer, and an explanation.	Yes - LLM
AI-Enhanced Cheating Detection	The system detects suspicious behaviour by monitoring tab switching and very fast answering, defined as an average of less than 5 seconds per question.	Rule-based / Heuristic

AI Workflow

The following flowchart describes how the selected course and teacher-entered topic move through the frontend, Laravel backend, course-aware prompt construction, AI service, response validation, and final review process.

AI Workflow – AI-Based Smart Quiz System



Step-by-step Explanation :

1. Teacher opens a course and enters a topic- The teacher first opens the required course, such as Database Management System, and enters a topic, such as Normalization, in the AI question-generation interface.

2. Frontend sends the course and topic to the backend- The React/Inertia frontend sends the current course ID together with the teacher-entered topic to the Laravel backend.

3. Laravel controller validates the request- The Laravel controller validates the course and topic, confirms that the selected course belongs to the teacher, and rejects invalid or empty input.

4. Backend constructs a course-aware AI prompt- Laravel loads the selected course name from the database and automatically builds a strict prompt using both the course name and topic. The prompt instructs the AI not to mix unrelated courses or add seed numbers, timestamps, or artificial context.

5. Backend calls the Groq API- The constructed course-aware prompt is sent to the Groq API, where the Llama 3.3 70B model processes the request and prepares an AI-generated response.

6. AI returns a structured JSON response- The model returns structured JSON containing the generated question, four options, the correct answer, and an explanation.

7. Laravel parses and validates the JSON response- The backend decodes the JSON and checks the required fields, exactly four non-empty unique options, the correct answer, and duplicate question text within the selected course.

8. The system checks whether the AI response is valid- If the response passes validation, the process continues to the question form. If the response is invalid, empty, duplicated, malformed, or the API request fails, the system moves to error handling.

9. Valid response: auto-fill the question form- The frontend receives the validated course-specific content and automatically fills the question, four options, correct answer, and explanation fields.

10. Invalid response: error handling and manual fallback- The system shows a user-friendly error message and keeps the manual question form available. It does not insert a dummy question from another course.

11. Teacher reviews, edits, and saves the question- The teacher reviews the generated question, edits it if necessary, and saves the approved question under the same selected course in the database.

Prompt Engineering

Prompt engineering is used to keep AI outputs course-specific, topic-specific, educational, and suitable for direct processing by the application. The backend automatically provides the selected course, teacher-entered topic, strict subject-isolation rules, the required JSON format, and answer-validation requirements.

System Prompt :

"You are an expert university-level educational MCQ generator. Generate exactly one academically accurate multiple-choice question strictly related to the selected course and selected topic. Do not mix unrelated courses or subjects, do not add seed numbers or timestamps, and return only valid JSON with four unique options, one exact correct answer, and a clear explanation."

User Input (Teacher Provides) :

The teacher enters only a topic in the AI question generator. The current course is taken automatically from the page. Example: Course = "Database Management System" and Topic = "Normalization".

Backend-Generated Prompt Sent to AI :

Laravel combines the current course name and teacher-entered topic and sends a course-aware instruction to the Groq API. For example:

"Selected Course: Database Management System. Selected Topic: Normalization. Generate one university-level MCQ strictly for this selected course and topic. Do not mix concepts from unrelated courses or subjects. Do not include seed numbers, timestamps, random numbers, or artificial context. Return ONLY valid JSON in the required structure:"
{"question":"...", "options":["A. ...", "B. ...", "C. ...", "D. ..."], "correct_answer":"A. ...",

"explanation":"..."}. The correct_answer must exactly match one of the four option strings.

Expected AI Output :

An example of the expected structured AI output for Database Management

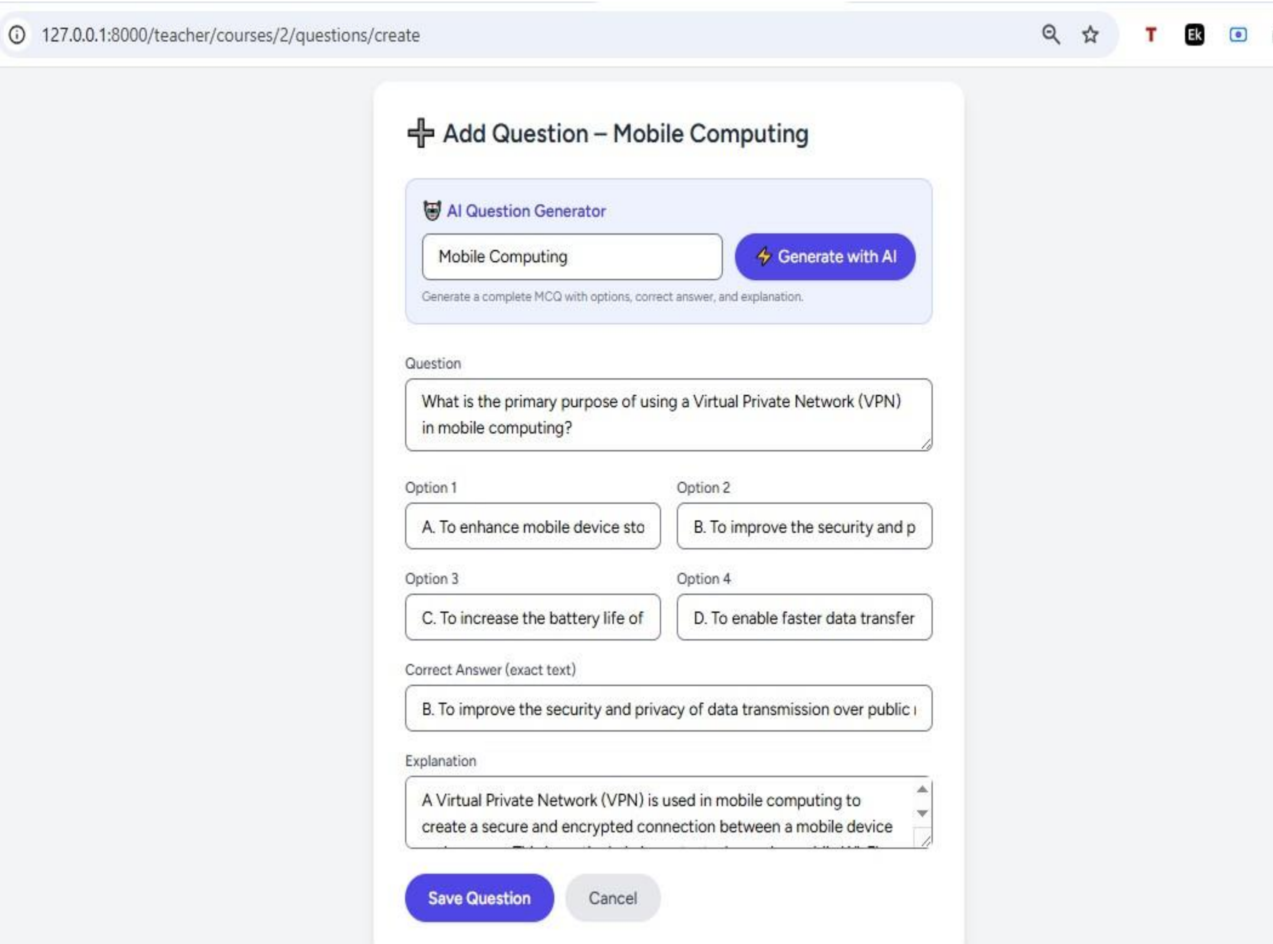
System → Normalization is shown below: json

```
{
  "question": "What is the primary purpose of normalization in a relational
  database?", "options": [
    "A. To reduce data redundancy and improve data integrity", "B.
    To increase duplicate data", "C. To remove all relationships
    between tables",
    "D. To store all data in a single table"
  ],
  "correct_answer": "A. To reduce data redundancy and improve data integrity",
  "explanation": "Normalization organizes relational data to reduce unnecessary duplication and
improve consistency and integrity."
}
```

Screenshots

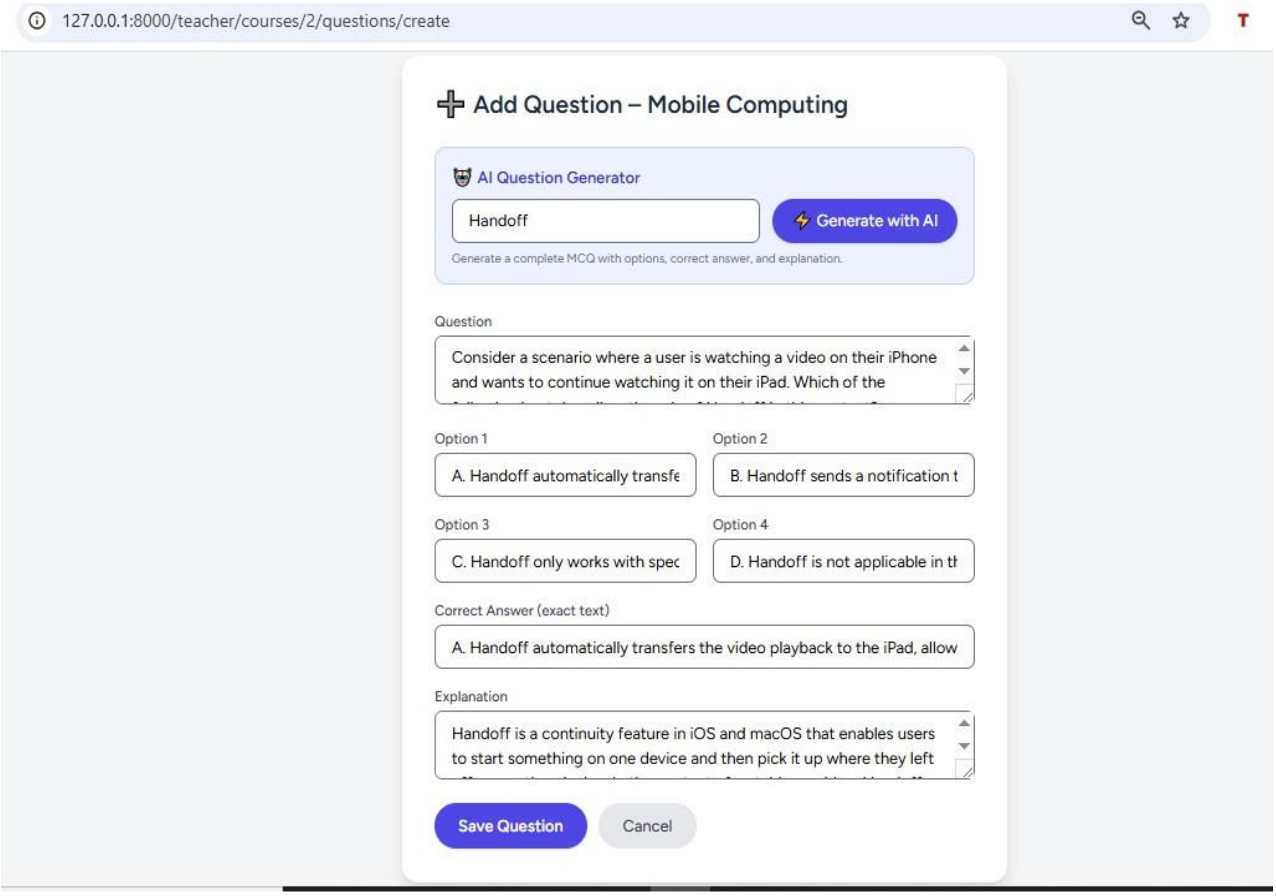
The following screenshots provide evidence of the implemented AI feature, real AI interaction, multi-course question generation, course-aware prompt input, backend integration, and repository activity.

Screenshot 1: Real AI Mode - Mobile Computing Topic



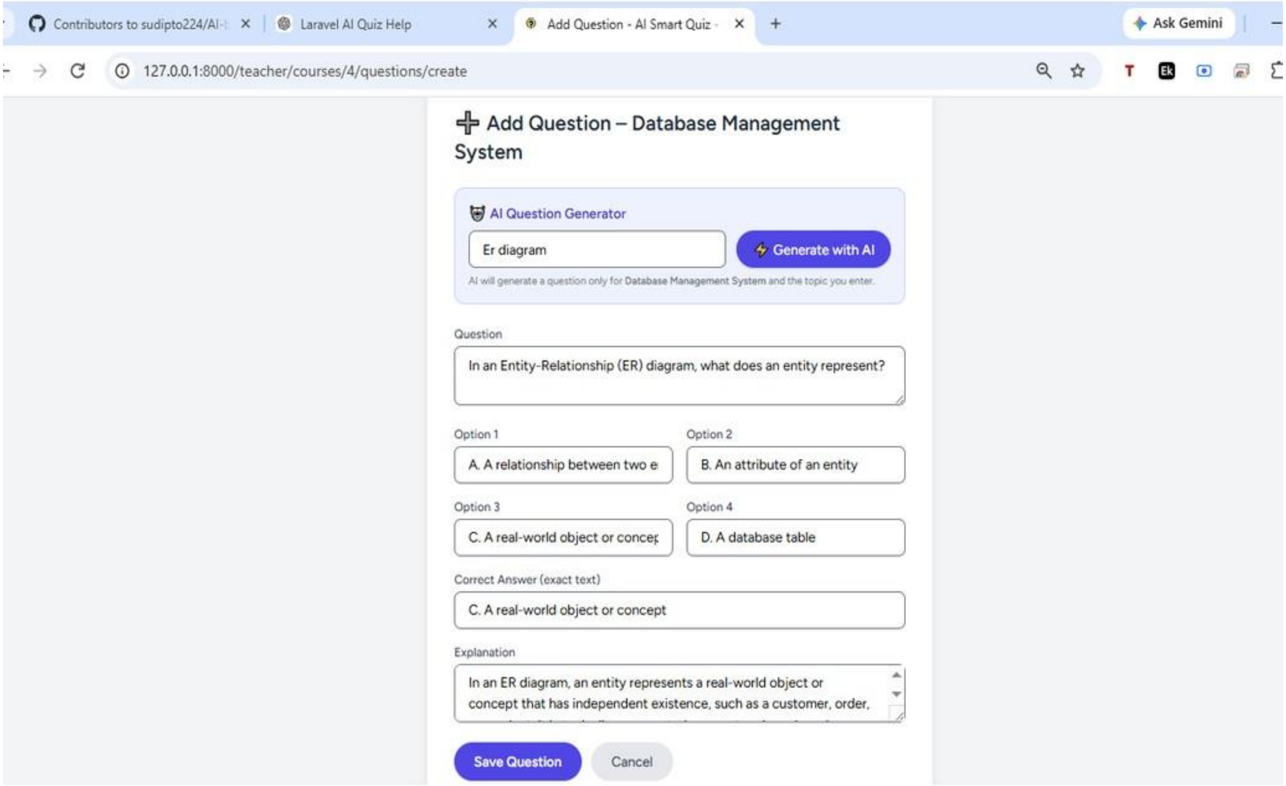
Description: This screenshot demonstrates the real-AI question-generation workflow in the teacher panel. After a topic is submitted, the form is automatically populated with a generated Mobile Computing MCQ, four options, the correct answer and an explanation, ready for review and saving.

Screenshot 2: Real AI Mode - Handoff Topic



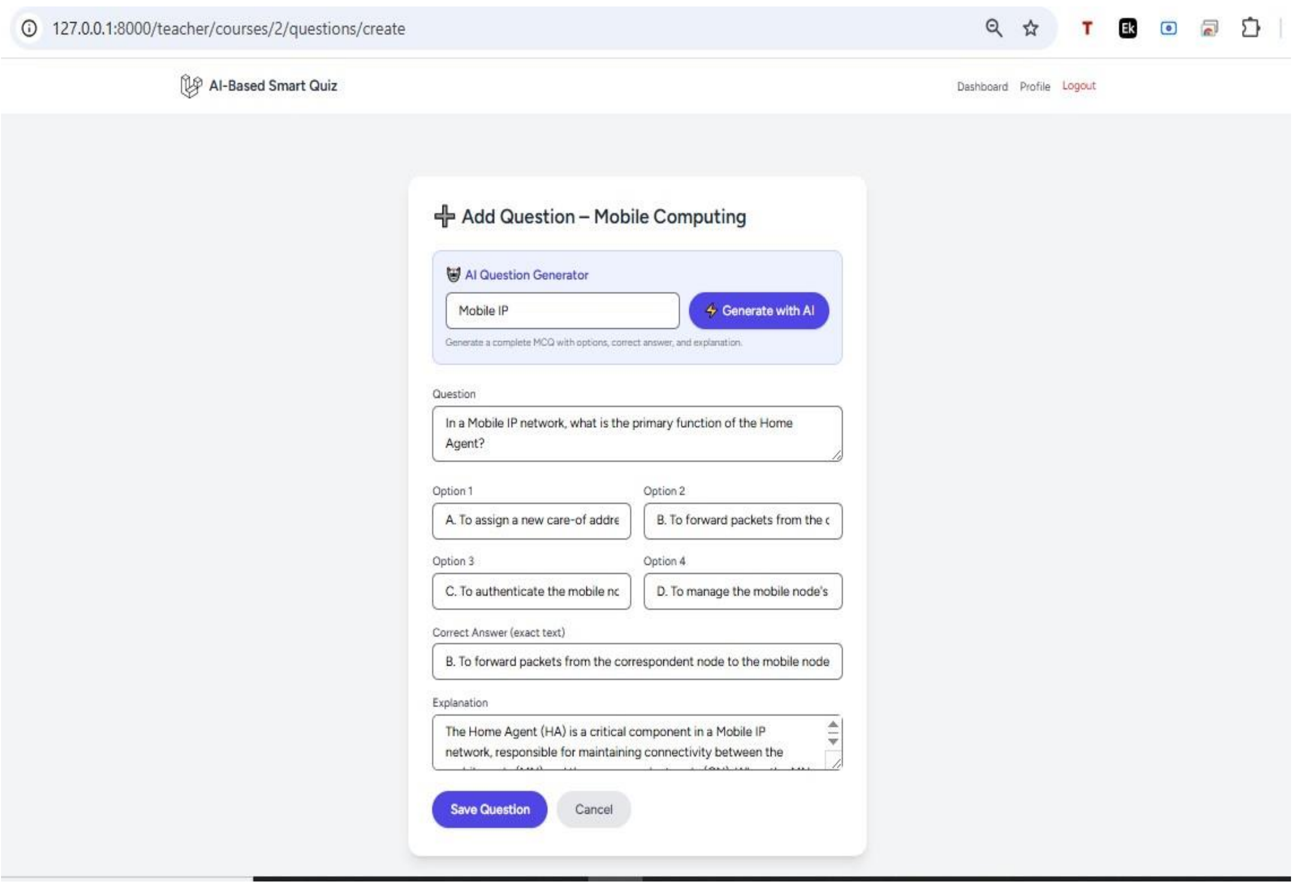
Description: This screenshot demonstrates successful real-AI generation for the Handoff topic. The Groq response is parsed and the question form is automatically populated with a topic-specific MCQ, answer and explanation.

Screenshot 3: Multi-Course AI Test - Database Management System



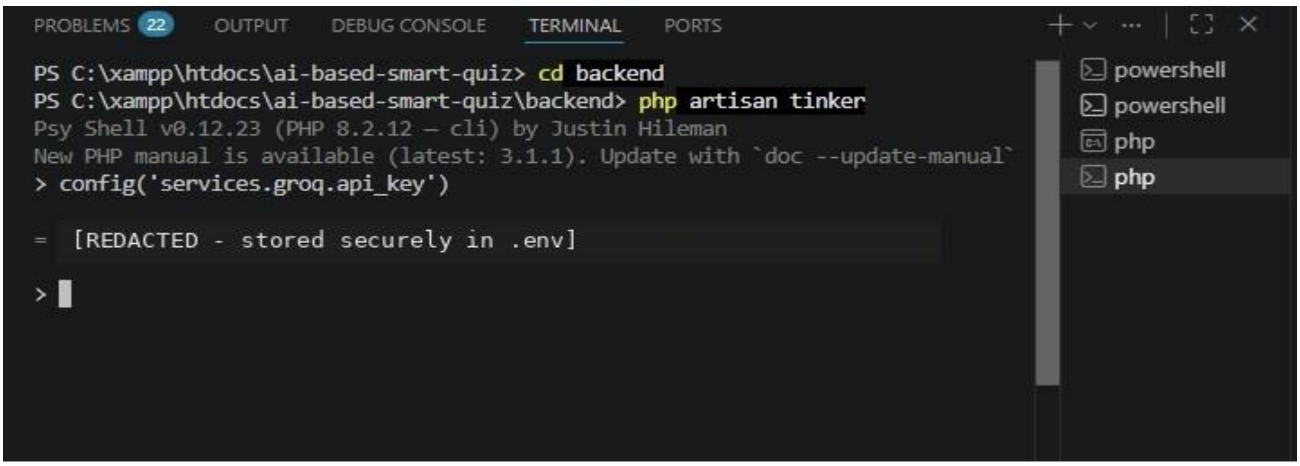
Description: This screenshot verifies the updated multi-course AI workflow. Inside the Database Management System course, the teacher enters the topic "ER diagram" and Groq generates a DBMS-specific Entity-Relationship question, four options, the exact correct answer, and an explanation without mixing Mobile Computing or unrelated context.

Screenshot 4: AI Interaction Interface (Teacher Panel)



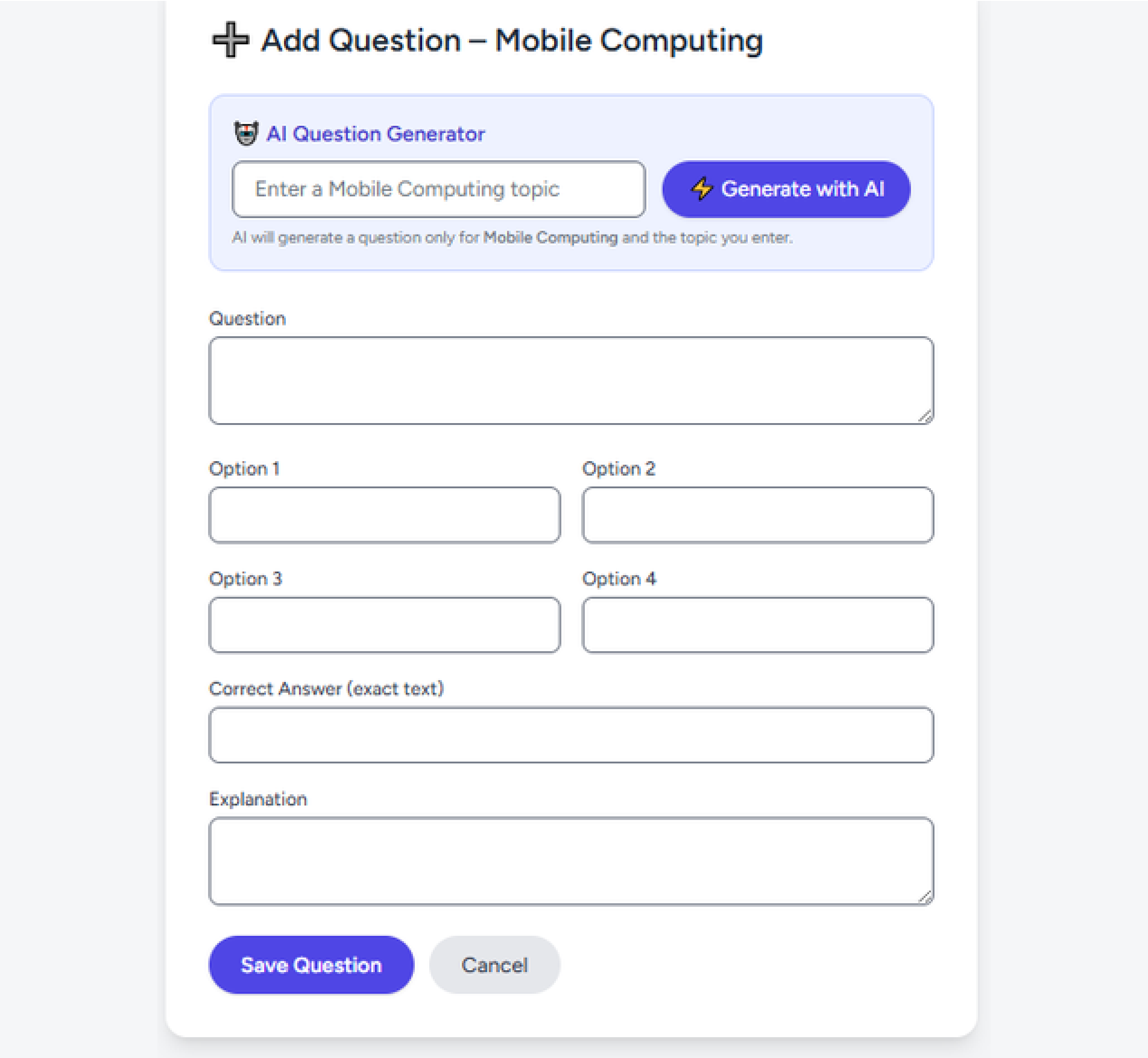
Description: This screenshot shows the teacher-facing AI Question Generator in the Mobile Computing course. A Mobile IP topic is submitted, and the generated question, four options, correct answer and explanation are displayed for teacher review before saving.

Screenshot 5: Backend Integration



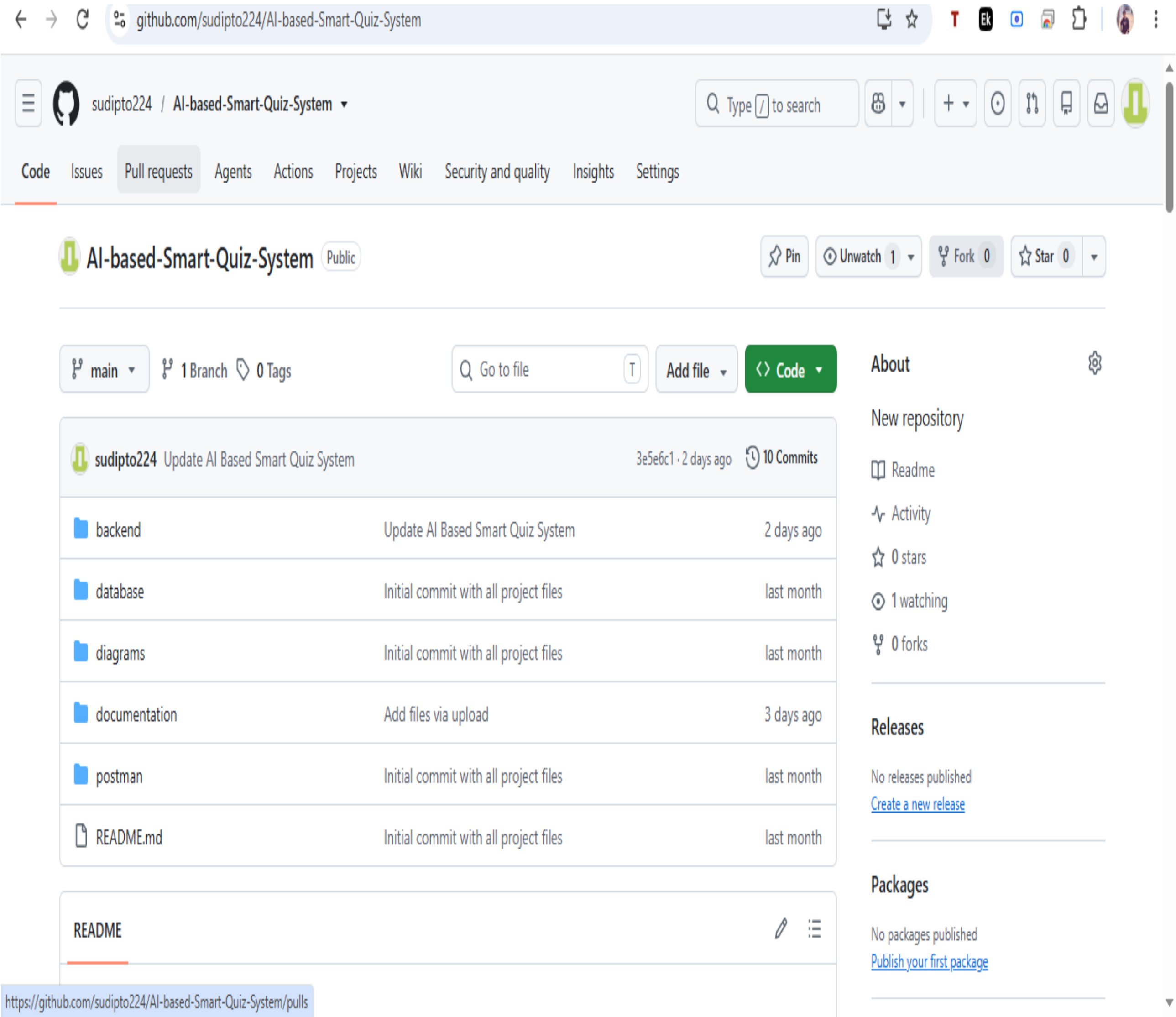
Description: This screenshot verifies the Laravel backend configuration used for the Groq integration. The application reads the Groq API key through the services configuration/environment setup.

Screenshot 6: Course-Aware AI Prompt Input Interface



Description: This screenshot shows the updated teacher-facing AI Question Generator. The current course name is displayed automatically, the topic field is course-aware, and the teacher only enters a topic. The frontend sends the current course ID and topic to Laravel, which constructs the strict course-specific prompt before calling Groq.

Screenshot 7: GitHub Repository



AI Response Handling Strategy

The system is designed to remain usable even when the AI service returns an invalid result or becomes temporarily unavailable. Errors are logged, a user-friendly message is returned, and the teacher can continue with the manual question form instead of receiving an unrelated dummy question.

Scenario	Handling Strategy
Successful Response	Parse and validate the JSON response, auto-fill the question form, and display a success message containing the selected course and topic.
Empty Response	Log the empty response and return a user-friendly retry message; the manual question form remains available.
Invalid JSON	Log the JSON parsing/validation error and return a user-friendly message; do not insert a dummy question.
API Error (4xx/5xx)	Log the API status and response body, then return a user-friendly service-unavailable/retry message.
Network Failure	Use a 30-second timeout. If the request fails, log the exception and keep manual question entry available.
Rate Limit (429)	Treat HTTP 429 as an API error, log the status, and return the standard user-friendly retry/manual-entry message.
Quota Exceeded	Treat quota-related API failure as an API error, log it, and return the standard user-friendly retry/manual-entry message.
Fallback Strategy	If Groq fails or returns invalid content, the system does not auto-fill a dummy question. It shows a user-friendly error and leaves the manual question form available so the teacher can continue safely.

AI Output Validation

To improve reliability, AI-generated content is validated before it is accepted into the question-creation workflow. The validation process checks both the structure and the educational usability of the returned content.

Validation Check	Validation Action
Empty Response	Reject an empty AI response, log the issue, and return a clear retry/manual-entry message.
Invalid JSON / Required Fields	Confirm valid JSON and require question, options, correct_answer, and explanation before using the response.
Formatting Cleanup	Remove unexpected Markdown code fences and surrounding whitespace before JSON processing.
Duplicate Question Prevention	Check generated question text against existing questions in the same selected course and reject exact duplicates.
Correct Answer Validation	Require exactly four unique non-empty options and ensure correct_answer exactly matches one of the option strings.

Current Development Progress

The Week 08 AI integration tasks have been implemented and tested in real AI mode, multi-course mode, course-aware prompt generation, validation, and error/manual-fallback handling. The current development status is shown below.

Component	Progress	Status
Groq API Integration	100%	Complete
Prompt Engineering	100%	Complete
AI Response Validation	100%	Complete
Error Handling & Logging	100%	Complete
Manual Fallback / Error Handling	100%	Complete

Frontend AI Interface	100%	Complete
Testing (Real + Multi-Course)	100%	Complete

Current Limitations

- The AI question generator depends on internet connectivity and Groq API availability.
- AI-generated questions may occasionally be inaccurate or ambiguous, so teacher review is required before saving.
- The course-aware prompt supports multiple courses, but the system still relies on the teacher to enter an appropriate topic and does not automatically verify that every entered topic truly belongs to the selected course.
- The cheating-detection component is rule-based; tab switching and very fast answering indicate suspicious behaviour but do not prove cheating.

Future Improvements

- Introduce adaptive difficulty based on each student's previous quiz performance.
- Generate personalised quizzes and practice questions from individual strengths and weaknesses.
- Provide AI-generated feedback and explanations for incorrect answers.
- Recommend weak topics automatically and suggest targeted practice.
- Add automatic course-topic relevance checking, support batch question generation, strengthen duplicate detection, and improve cheating analysis.

GitHub Repository

The project source code and ongoing development history are maintained in the team GitHub repository.

Repository Link: https://github.com/sudipto224/AI-based-Smart-Quiz-System